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## The following core functionality is supported

- Hardware abstraction layer (HAL)
- Real-time kernel
- Interrupt handling
- Exception handling
- Choice of schedulers
- Thread support
- Rich set of synchronisation primitives
- Timers, counters and alarms
- Choice of memory allocators
- Debug and instrumentation support
- $\mu$ ITRON 3.0-compatible API
- POSIX-compatible API
- ISO C and Maths libraries
- Serial, Ethernet, SPI, I2C, framebuffer, CAN, ADC, wallclock and watchdog device drivers
- USB slave support
- TCP/IP networking stacks
- C++ standard template library (uSTL)
- GDB debug support

for a fast response to external events, you cannot rely on Linux; you would need to use eCos.

## What is configurability

As mentioned in the first line, the configurability part of eCos is one of its biggest selling points. Being able to configure your own OS according to application enables developers to fully utilise the power of the OS to optimise performance, and not give in to system demands (which are almost always different from developer needs). eCos allows the developer/application writer to impose the requirements, both in terms of functionality and requirements, on to run-time components.

One of the extremely important reasons configurability increases the relevance of eCos is that this way it removes all functionalities that are not essential to the application, which drives down the resource footprint of eCos. Imagine MotoGP, with all their bikes, essentially containing only the necessary propulsion components. Their only concern is being the fastest while making sure the bike finishes the race. The bikes are

a stripped-down and optimised version of their street avatars much like eCos. Add to that the completely-free nature of eCos, which means you do not have to pay a single paisa for the development/deployment of your application. May be it is possible to have your cake and eat it, too.

Configurability allows one to divide the various packages into components. You can separate it into core packages and non-core packages. Core packages take care of the kernel, C library and Maths library, while additional ones are added depending on specific applications. Suppose for sending data, you want to add on a TCP/IP stack. Now, the TCP/IP stack has certain dependencies that can be added later on.

## Where do I sign up

eCos has several mailing lists for the interested, some for announcements, some for releases and some for general queries. Rest assured, any and all doubts that you have will be addressed by the crowd of techies already deep into development with eCos. As of today, eCos supports the following architectures:

- 68K/ColdFire
- ARM (ARM7TDMI, ARM9TDMI, Cortex-M, StrongARM and Xscale)
- CalmRISC16 and CalmRISC32
- Fujitsu FR-V
- Fujitsu FR30
- Hitachi H8/300
- Intel x86
- Matsushita AM3x
- MIPS
- NEC V8xx
- PowerPC
- SPARC
- SuperH

We have bundled the latest version in this month's DVD, so you can try for yourself the power of configurability and let your own contribution fly into the connected world of the future. **EFY**

*The author is a student at Great Lakes Institute of Management (GLIM), Chennai*